TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Advanced Testing and Validation Platforms for High-Performance AEM Electrolysis Systems Targeting European Market Entry*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Knowledge Services*	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	*Hydrogen*	*R&D / Engineering Services*	*Testing / Inspection / Analysis Techniques*
Concept of Proposed Technology	The proposed technology aims at developing an advanced, modular testing and validation platform for AEM water electrolysis cells and short stacks to accelerate industrial qualification and European market entry. The platform combines Fraunhofer IFAM's electrochemical diagnostics and materials degradation know-how with industrial-grade system engineering to enable standardized benchmarking, long-term durability testing, and accelerated stress testing under controlled temperature, pressure, and load profiles. Key functions include automated test execution, traceable data acquisition (electrochemical, thermal, fluidic and gas-quality), and diagnostic workflows linking performance losses to component degradation. Potential uses include R&D qualification, quality assurance, customer acceptance testing, and preparation of documentation supporting conformity assessment and CE marking of commercial AEM electrolyzer products in Europe.			

(max 10 pages)

Rationale for Joint R&D Project	The joint R&D project is needed to close the gap between electrochemical insight and industrially robust validation infrastructure for AEM electrolysis. Fraunhofer IFAM contributes deep expertise in electrochemical technology, development of meaningful test protocols, advanced diagnostics, and materials/component degradation analysis from cell level up to short-stack testing. The Korean SME partner complements this with strong system engineering capabilities to design, build, automate and operate scalable testing equipment and to translate laboratory requirements into a manufacturable, safe prototype. Collaboration is essential because credible European market validation requires both scientifically sound degradation understanding and industrial-grade test platform implementation, enabling reliable, reproducible data for customer qualification and conformity assessment support.
Key Words	1. European Market Entry & Conformity Assessment 2. Industrial AEM Electrolysis 3. Cell & Short-Stack Test Platforms 4. Durability and Accelerated Stress Testing 5. Electrochemical Degradation Diagnostics

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	The objective of this R&D project is to develop and validate a prototype AEM electrolysis testing platform and associated testing methodologies that enable reliable performance and durability assessment and support European market entry. Specifically, the project aims to (i) design a modular test system for cells and short stacks including automation, safety concept, and a traceable measurement chain; (ii) develop and validate standardized protocols for performance, durability, and accelerated stress testing, including gas crossover/purity monitoring; (iii) demonstrate reproducible operation across an industrially relevant operating window and verify measurement repeatability; (iv) establish diagnostic workflows combining in-operando electrochemical analysis and post-mortem characterization to link performance losses to component degradation; and (v) provide a documentation package (test plans, reports, calibration/traceability, and risk assessment inputs) to support European customer qualification and preparation for CE marking of commercial AEM electrolysis systems.			
R&D Target Specifications	The project targets the development and validation of a prototype AEM electrolysis testing platform at TRL 5, enabling standardized and reproducible assessment of cells and short stacks. The testing equipment shall support active areas ≥ 100 cm^2 per cell and short stacks up to 5 cells, operating temperatures up to 80 °C and operating pressures up to ≥ 20 bar. It shall enable programmable load profiles up to ≥ 2.0 A cm^{-2} and provide synchronized measurement of per-cell voltage, current, temperatures, pressures, flow rates and online gas-quality/crossover indicators. The measurement chain shall be designed for traceability and calibrated operation, targeting ≤ ± 2 % repeatability of key KPIs (e.g., cell voltage at defined operating points and hydrogen production rate) across repeated test runs. The platform shall enable continuous durability testing for ≥ 1,000 h as well as accelerated stress testing with defined load/temperature/pressure cycles and automated safety shutdown procedures. The resulting datasets and reports shall be suitable for benchmarking, degradation analysis and for supporting conformity assessment documentation relevant to European market entry.			
R&D Plan	The project follows a phased, milestone-driven approach consistent with the program structure (Phase 1: 6-month pre-study; Phase 2: 30-month main R&D). Phase 1 will de-risk the development by defining user requirements, safety/compliance needs for European deployment, and validating the core test methodology at lab scale. The phase will conclude with a consolidated specification and an implementation-ready engineering package that also serves as input for the competitive Phase 2 application. Phase 2 will engineer, build and commission the prototype testing platform, execute a structured validation and long-term test campaign, and deliver standardized datasets and documentation supporting European customer qualification and conformity assessment. Phase 1: Pre-study (Months 1–6) WP1: Requirements, Safety and Compliance Mapping (Months 1–2) • Define functional/technical requirements (operating window, scalability, data quality)			

	• Preliminary hazard analysis and safety concept; mapping of CE-relevant directives/standards Deliverable D1: User Requirements Specification & Safety/Compliance Concept WP2: Testing Methodology and Data Quality Concept (Months 2–4) • Develop standardized performance, durability and AST protocols • Define measurement architecture, calibration needs and data quality/repeatability criteria Deliverable D2: Draft Test Protocol Suite & Data Quality Plan WP3: Laboratory Feasibility Demonstration and Phase-2 Preparation (Months 4–6) • Validate key measurement concepts using representative AEM cells at lab scale • Finalize prototype specification and preliminary design; prepare Phase-2 implementation plan Deliverable D3: Prototype Specification, Preliminary Design Package & Phase-2 Proposal Inputs Milestone M1 (end of Month 6): Validated concept and finalized specification enabling prototype development Phase 2: Main R&D Phase (Months 7–36) WP4: Prototype Engineering and Manufacturing (Months 7–14) • Detailed mechanical, electrical and control design; procurement and manufacturing • Implementation of safety functions and modular interfaces for cell/short-stack testing Deliverable D4: Prototype Test Platform Assembled WP5: Integration, Commissioning and Calibration (Months 15–20) • Integration of sensors, gas analysis and data acquisition; commissioning under defined conditions • Establish calibration procedures and verify measurement repeatability Deliverable D5: Commissioned Platform & Verified Measurement Chain WP6: Validation, Long-Term Testing and Degradation Diagnostics (Months 21–32) • Execute durability tests (≥ 1,000 h) and accelerated stress tests; collect standardized datasets • Perform in-operando diagnostics and post-mortem analysis to link performance losses to degradation Deliverable D6: Validated Test Results, Degradation Insights and Benchmark Dataset WP7: Documentation and Exploitation Preparation (Months 33–36) • Prepare test reports and documentation supporting conformity assessment and customer qualification • Define exploitation/commercialization pathway for the testing platform (product and/or service) Deliverable D7: Documentation Package & Exploitation Plan Milestone M2 (end of Month 36): TRL 5 prototype demonstrated with validated long-term testing

(max 10 pages)

	capability and documentation package

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer IFAM will provide scientific and technical leadership in electrochemical testing, diagnostic method development, and degradation analysis for AEM electrolysis. IFAM will define and validate performance, durability and accelerated stress testing protocols, including data quality and repeatability criteria, and will carry out advanced electrochemical diagnostics and post-mortem characterization (e.g., microscopy and chemical/phase analysis) to understand component degradation. IFAM will contribute its electrochemical laboratories and test infrastructure (cell/short-stack testing, online gas analysis and high-pressure hydrogen facilities) and will support independent validation, data interpretation and the definition of procedures aligned with European customer qualification and conformity assessment needs.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for the engineering, development and implementation of the prototype AEM electrolysis testing equipment and its automation. This includes mechanical and electrical system design, integration of control, safety and data acquisition systems, and fabrication/assembly of an industrially relevant test platform. The SME will lead prototype build-up, system integration and operational testing, and will execute long-term test campaigns together with IFAM, providing practical feedback to refine the equipment and protocols. Furthermore, the SME will ensure scalability and manufacturability of the developed platform and will lead exploitation and commercialization activities, including preparation for European market entry of both the SME's electrolyzer systems and the developed testing solution in close collaboration with Fraunhofer IFAM.

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will deliver a validated prototype AEM electrolysis testing platform at TRL 5 together with standardized test protocols and a traceable, repeatable measurement chain for cell and short-stack qualification. Technological outcomes include (i) a modular test system enabling long-term durability testing and accelerated stress testing under industry-relevant temperature and pressure conditions, and (ii) diagnostic workflows that correlate electrochemical signatures with materials/component degradation, generating actionable insights for design improvement and reliability assessment. Commercially, the prototype platform and validation methodology will reduce time, cost and uncertainty for European customer qualification by providing credible performance and lifetime evidence and documentation supporting conformity assessment and CE marking preparation. At a broader industrial and economic level, the project will strengthen the competitiveness and export readiness of Korean hydrogen technology suppliers, enable new testing and validation service/product offerings, and reinforce long-term Germany-Korea cooperation with Fraunhofer IFAM in industrial hydrogen technologies.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

(max 10 pages)